

Chapter

Sound

1. Two sound waves X and Y have the same amplitude and the same wave pattern but their frequencies are 60 Hz and 120 Hz respectively, then

- (a) X will be shriller and Y will be grave
- (b) X will be grave and Y will be shriller
- (c) X will differ in quality than Y
- (d) X is louder than Y.

SQP2025

2. (a) Name the waves used for echo depth sounding.

(b) Give one reason for their use in the above purpose.

SQP 2025

3. A person standing in front of a cliff fires a gun and hears its echo after 3 s. If the speed of sound in air is 336 ms^{-1} .

(a) Calculate the distance of the person from the cliff.

(b) After moving a certain distance from the cliff, he fires the gun again and this time the echo is heard 1.5 s later than the first. Calculate the distance that the person moved.

SQP 2025

4. When the stem of vibrating tuning fork is pressed on a table, the tabletop starts vibrating. These vibrations are definitely an example of

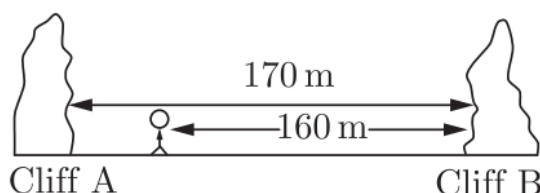
- (a) Resonance
- (b) Natural vibrations
- (c) Forced vibrations
- (d) Damped vibrations

MAIN 2024

5. Rohan conducted experiments on echo in different media. He observed that a minimum distance of ' x ' metres is required for the echo to be heard in oxygen and ' y ' metres in benzene. Compare ' x ' and ' y '. Justify your answer.

MAIN 2024

6. (a) Name the waves used in SONAR.



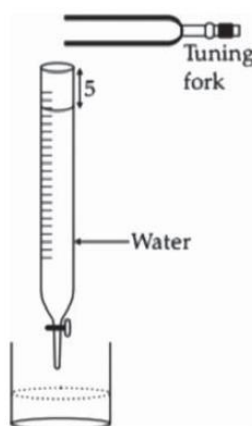
(b) In the above diagram Lata stands between two cliffs and claps her hands.

Determine the time taken by her to hear the first echo.

Speed of sound in air 320 ms^{-1} .

MAIN 2024

7. In the given diagram, a vibrating tuning fork is kept near the mouth of a burette filled with water. The length of the air column is adjusted by opening the tap of the burette. At a length of 5 cm of the air column, a loud sound is heard.



(a) Name the phenomenon illustrated by the above experiment.

(b) Why is a loud sound heard at this particular length?

(c) If the present tuning fork is replaced with a tuning fork of higher frequency, should the length of the air column increase or decrease to produce a loud sound? Give a reason.

MAIN 2024

8. Name any two electromagnetic waves which have a frequency higher than that of violet light. State one use of each.

SQP 2023

9. A tuning fork (vibrating) is held close to ear. One hears a faint hum. The same (vibrating) tuning fork is placed on table, such that its handle is in contact with table, one hears a loud sound. Explain.

MAIN 2023

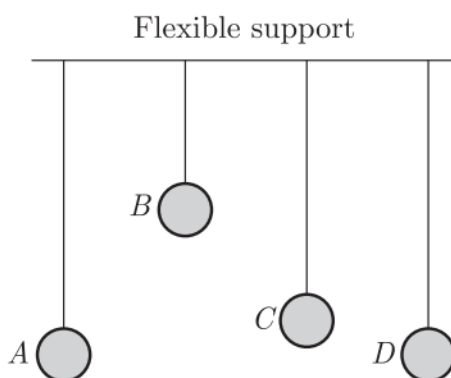
10. A radar sends a signal to an aircraft at a distance of 30 km away and receives it back after 2×10^{-4} second. What is the speed of the signal?

MAIN 2023

11. How do you account for the fact that two strings can be used to give notes of the same pitch and loudness but of different quality?

COMP 2023

12. In the given figure pendulum (A) is set into vibrations, which of the other three will vibrate with maximum amplitude and why?



COMP 2023

13. A man standing 25 m away from a wall produces a sound and receives the reflected sound.

(a) Calculate the time after which he receives the reflected sound if the speed of sound in air is 350 ms^{-1} .

(b) Will the man be able to hear a distinct echo? Give a reason for your answer.

COMP 2023

14. A man stands in between two high rise buildings and blows a whistle. He hears two successive echoes after 0.4 s and 1.6 s. Calculate the distance between buildings [Speed of sound = 332 ms^{-1}]

- (a) 132 m (b) 332 m
(c) 232 m (d) 432 m

SQP 2022

15. (i) State the safe limit of sound level in terms of decibel for human hearing.

(ii) Name the characteristic of sound in relation to its waveform.

SQP 2022

16. A person standing between two vertical cliffs and 640 in away from the nearest cliff shouted. He heard the first echo after 4 seconds and the second echo 3 seconds later. Calculate (i) the velocity of sound in air and (ii) the distance between the cliffs.

SQP 2022

17. Echo occurs when the distance between source and listener is

- (a) 17 m (b) greater than 17 m
(c) 10 m (d) 34 m

MAIN 2022

18. The pressure amplitude in sound wave from a radio receiver is $2.0 \times 10^{-2} \text{ N/m}^2$ and the intensity at a point is $5.0 \times 10^{-7} \text{ W/m}^2$. If by turning the volume knob, the pressure amplitude increased to $2.5 \times 10^{-2} \text{ N/m}^2$. Evaluate the intensity.

- (a) $7.8 \times 10^{-7} \text{ W/m}^2$ (b) $7.8 \times 10^{-6} \text{ W/m}^2$
(c) $7.8 \times 10^{-8} \text{ W/m}^2$ (d) 7.8 W/m^2

MAIN 2022

19. What do you understand by the terms ?

(i) Wavelength, (ii) Frequency and (iii) Amplitude of a wave.

MAIN 2022

20. A man stands at a distance of 68 m from a cliff and fires a gun. After what time interval will he hear the echo, if the speed of sound in air is 340 m s^{-1} ?

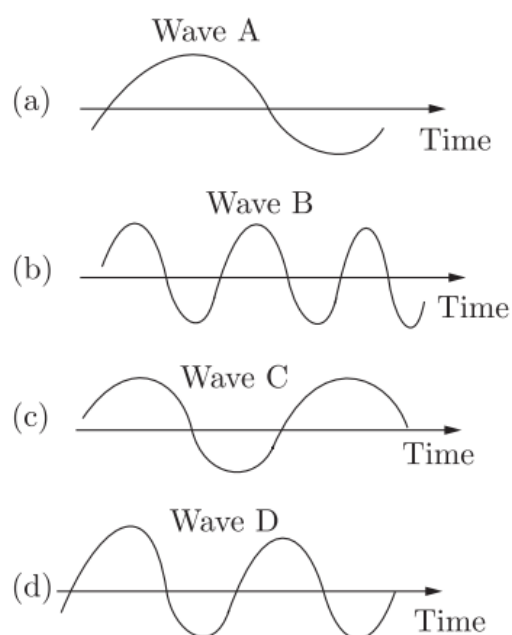
MAIN 2022

21. Free vibrations are

- (a) The vibrations under the influence of a periodic force
- (b) The vibrations with larger amplitude
- (c) The vibrations when the frequency continuously decreases
- (d) The vibrations with a constant frequency and constant amplitude

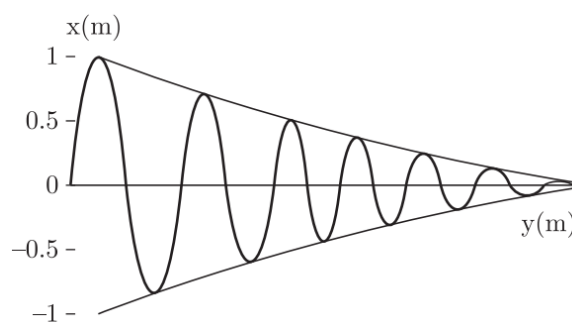
SEM-II 2022

22. The diagram below shows four sound waves. Which sound has the highest pitch?



SEM-II 2022

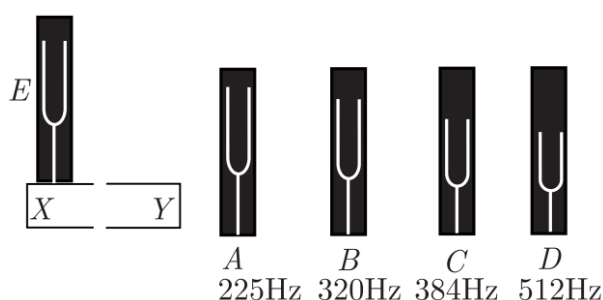
23. Study the below figure and answer the following:



- (i) What type of vibration does the above figure represent?
- (ii) State one reason for which the amplitude of the vibration decreases with time.
- (iii) Write an example of natural vibrations.

SEM-II 2022

24. The diagram below shows a vibrating tuning fork *E* mounted on a sound box *X*. When the vibrating tuning forks *A*, *B*, *C* and *D* are placed on the sound box *Y* one by one, it is observed that a louder sound is produced when the tuning fork *B* is placed on *Y*.



- (i) What is the frequency of tuning fork *E* ?
- (ii) Why does *B* produce a louder sound?

SEM-II 2022

25. Define pitch of a note. What are the factors upon which pitch of a note depends?

SQP 2021

26. Define the terms:

- (i) Amplitude
- (ii) Frequency (as applied to sound waves)

MAIN 2021

27. Why an echo is not heard when distance between source of sound and reflecting body is 10 m?

MAIN 2021

28. Which one of the following diagnostic methods use reflection of sound?

- (a) CT scan
- (b) Electrocardiogram
- (c) Echocardiogram
- (d) MRI

SEM-I 2021

29. Select a correct option with respect to echo depth sounding.

- (a) Infrasonic waves are used.
- (b) The frequency of the waves used is between 20 Hz and 20000 Hz.
- (c) Ultrasonic waves are used.
- (d) Supersonic waves are used.

SEM-I 2021

30. A man stands in between two cliffs, such that he is at a distance of 133.6 m from nearer cliff. He fires a gun and hears first echo after 0.8 s and second echo after 1.8 s. The distance between two cliffs is

- (a) 434.2 m
- (b) 234.2 m
- (c) 334.2 m
- (d) 134.2 m

COMP 2021

31. Why does rear view mirror of a motor bike start vibrating violently, at some particular speed of motor bike?

COMP 2021

32. A certain sound has a frequency of 256 hertz and a wavelength of 1.3 m.

- (i) Calculate the speed with which this sound travels.
- (ii) What difference would be felt by a listener between the above sound and another sound traveling at the same speed, but of wavelength 2.6 m ?

COMP 2021

Solution

1.

Ans: (b) X will be grave and Y will be shriller

Explanation:

The **pitch (shrillness)** of a sound depends on its **frequency** — higher frequency gives a **shriller** sound, while lower frequency gives a **graver** sound.

For the given waves:

$$f_X = 60 \text{ Hz}, \quad f_Y = 120 \text{ Hz}$$

Since $f_Y > f_X$, wave **Y** is **shriller**, and wave **X** is **grave**.

So, sound wave **X** will be grave and **Y** will be shriller, which matches **option (b)**.

2.

Ans: (a) Ultrasonic waves are used for echo depth sounding.

(b) They are used because **ultrasonic waves can travel long distances without much loss of energy**, are **not easily absorbed by the medium**, and **can be confined to a narrow beam**, making them suitable for accurate depth measurement.

3.

Ans:

(a)

$$d = \frac{s \times t}{2}$$

$$d = \frac{336 \times 3}{2}$$

$$d = 504 \text{ m}$$

(b) New echo time = 1.5 s

$$d = \frac{336 \times 1.5}{2}$$

$$d = 252 \text{ m}$$

So, the **distance the person moved**

$$= 504 - 252 = 252 \text{ m}$$

4.

Ans: (c) Forced vibrations

Explanation:

When the vibrating tuning fork is pressed on the table, it applies a **periodic external force** on the tabletop.

Due to this external force, the table is made to vibrate, even though it was not vibrating on its own. Such vibrations produced by an **external periodic force** are called **forced vibrations**.

5.

Ans: To hear an echo, the sound must travel a minimum distance of **17 metres in oxygen** and **10 metres in benzene**.

Using the formula:

$$d = v \times \frac{0.1}{2}$$

For oxygen:

$$x = 340 \times \frac{0.1}{2} = 17 \text{ m}$$

For benzene:

$$y = 200 \times \frac{0.1}{2} = 10 \text{ m}$$

Therefore,

$$x : y = 17 : 10$$

So, the minimum distance required for hearing an echo is greater in oxygen than in benzene because the speed of sound is higher in oxygen.

6.

Ans: (a) Ultrasonic waves.

(b) Lata cannot clearly hear the echo from **Cliff A** because the distance (170 m)

is less than the minimum distance required to hear a distinct echo.

Therefore, she will hear the first echo from **Cliff B**, which is 160 m away.

$$t = \frac{2d}{v}$$

$$t = \frac{2 \times 160}{320} = 1 \text{ s}$$

7.

Ans: (a) Resonance.

(b) A loud sound is heard because the natural frequency of the air column becomes equal to the frequency of the tuning fork.

Hence, resonance occurs.

(c) The length of the air column should **decrease**.

A higher frequency requires a shorter air column because the wavelength is smaller, and the resonant length is directly proportional to wavelength (and inversely proportional to frequency).

8.

Ans: (i) Ultraviolet rays – Used for purifying water / detecting adulteration in ghee.

(ii) X-rays – Used for detecting fractures in bones.

9.

Ans: When a vibrating tuning fork is held near the ear, it sets only a small amount of air into vibration, so the sound heard is faint.

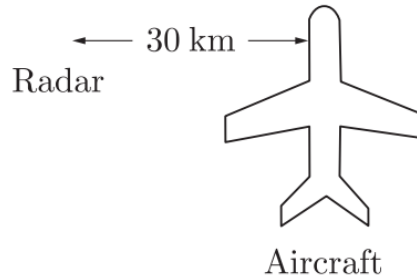
But when its handle touches the table, the vibrations are passed on to the table, causing **forced vibrations**.

Because the table has a large surface area, it vibrates a large volume of air, producing a **much louder sound**.

10.

Ans: Speed of the signal sent to an aircraft,

$$\begin{aligned}\nu &= \frac{2d}{t} \\ &= \frac{2 \times 30 \times 1000 \text{ m}}{2 \times 10^{-4} \text{ s}} \\ &= 3 \times 10^8 \text{ m s}^{-1}\end{aligned}$$



11.

Ans: The **quality** or **timbre** of a sound depends on the **harmonics** present in it.

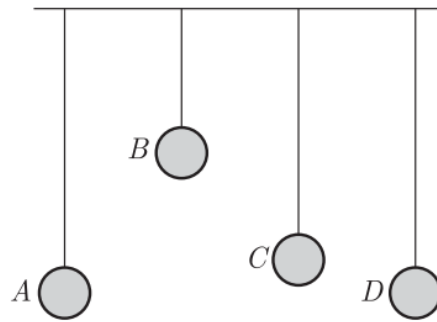
Even if two strings produce notes of the **same pitch** (same fundamental frequency) and **same loudness**, the **harmonics** produced by each string may be **different**.

Because the **mixture and relative strengths of harmonics differ**, the two notes will have **different quality**, even though their fundamental frequencies are the same.

12.

Ans: According to the shown figure the pendulums (A) and (D) are of the same length, therefore, their frequencies of the vibrations are the same. Now if pendulum (A) is displaced from its mean position, let it performs natural vibrations which causes the flexible support to vibrate with the frequency of (A). Through the support, these vibrations are transferred to the other pendulums due to which other pendulums also begin to vibrate with the natural frequency of (A).

Pendulums (B) and (C) vibrate with smaller amplitude than that of (A) but with the same frequency as that of (A). Pendulums (B) and (C) are in the state of forced vibrations. Pendulum (D) also starts vibrating and ultimately acquires the same amplitude as that of (A). Pendulum (D) being of the same length attains the state of resonance.



13.

Ans: (a) Distance to the wall = 25 m

Total distance travelled by sound = $2 \times 25 = 50$ m

Speed of sound = 350 m s^{-1}

$$t = \frac{2d}{v}$$

$$t = \frac{2 \times 25}{350} = 0.14 \text{ s}$$

So, the man receives the reflected sound after **0.14 s**.

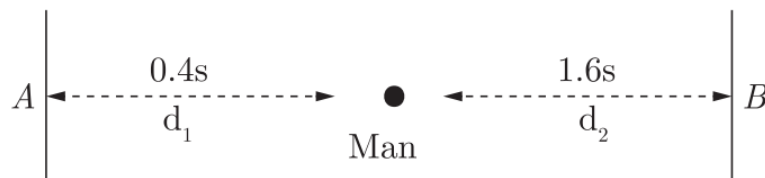
(b) Yes, he will hear a distinct echo because the time interval **0.14 s** is greater than **0.1 s**, which is the minimum time required to hear an echo.

14.

Ans:

Let the two buildings **A** and **B** be at distances

d_1 and d_2 from the man.



For the **first echo**:

$$t = 0.4 \text{ s}$$

$$d_1 = \frac{vt}{2} = \frac{332 \times 0.4}{2} = 66.4 \text{ m}$$

For the **second echo**:

$$t = 1.6 \text{ s}$$

$$d_2 = \frac{vt}{2} = \frac{332 \times 1.6}{2} = 265.6 \text{ m}$$

Total distance between the buildings:

$$AB = d_1 + d_2 = 66.4 + 265.6 = 332.0 \text{ m}$$

Therefore, the correct option is **(b) 332 m**.

15.

Ans: (i) Safe limit of sound level for human hearing is upto 120 dB.

(ii) Quality is that subjective characteristic of sound which enables us to distinguish between two sounds of the same loudness and same pitch but emitted by two different sources.

16.

Ans: (i) For the first echo, the total time heard is **4 s**.

Since the sound travels to the cliff and back, the time taken to reach the nearest cliff is:

$$t = \frac{4}{2} = 2 \text{ s}$$

Given distance to the nearest cliff = **640 m**

So, the velocity of sound is:

$$v = \frac{S}{t} = \frac{640}{2} = 320 \text{ m/s}$$

(ii) The second echo is heard **3 seconds after** the first echo.

So, the total time for the second echo =

$$4 + 3 = 7 \text{ s}$$

Time taken to reach the second cliff:

$$t = \frac{7}{2} = 3.5 \text{ s}$$

Distance to the second cliff:

$$x = v \times t = 320 \times 3.5 = 1120 \text{ m}$$

Therefore, the distance between the two cliffs is:

$$D = 640 + 1120 = 1760 \text{ m}$$

17.

Ans: (b) greater than 17 m

Explanation:

To hear an echo, the reflected sound must reach the ear **at least 0.1 s** after the original sound.

Speed of sound in air $\approx$ **340 m/s**.

Distance travelled in 0.1 s:

$$\text{distance} = 340 \times 0.1 = 34 \text{ m}$$

This 34 m is the **total** distance (to the wall and back).

So, the minimum distance between the listener and the reflecting surface is:

$$\frac{34}{2} = 17 \text{ m}$$

Therefore, for an echo to be heard, the distance between source and listener must be **greater than 17 m**.

18.

Ans: (a) $7.8 \times 10^{-7} \text{ W/m}^2$

Explanation:

For a sound wave, the **intensity** I is proportional to the **square of the pressure amplitude** p :

$$I \propto p^2$$

So, for two situations:

$$\frac{I_2}{I_1} = \frac{p_2^2}{p_1^2}$$

Given:

$$p_1 = 2.0 \times 10^2 \text{ N m}^{-2},$$

$$I_1 = 5.0 \times 10^{-7} \text{ W m}^{-2},$$

$$p_2 = 2.5 \times 10^2 \text{ N m}^{-2}.$$

$$\frac{I_2}{I_1} = \left(\frac{2.5}{2.0} \right)^2 = (1.25)^2 = 1.5625$$

$$I_2 = 1.5625 \times 5.0 \times 10^{-7} \approx 7.8 \times 10^{-7} \text{ W m}^{-2}$$

Hence, the new intensity is

$$I_2 \approx 7.8 \times 10^{-7} \text{ W m}^{-2},$$

so, option **(a)** is correct.

19.

Ans:

(i) Wavelength:

The wavelength of a wave is the **linear distance covered by one complete wave**, or the distance between two consecutive crests or troughs.

(ii) Frequency:

Frequency is the **number of vibrations or oscillations made by a vibrating particle in one second**.

(iii) Amplitude:

Amplitude is the **maximum displacement of a vibrating particle from its mean (rest) position**.

20.

Ans:

Given:

Distance from cliff, $d = 68 \text{ m}$

Speed of sound, $v = 340 \text{ m s}^{-1}$

To hear an echo, the sound must travel to the cliff and back.

$$t = \frac{2d}{v}$$

$$t = \frac{2 \times 68}{340} = 0.4 \text{ s}$$

Time interval = 0.4 s

21.

Ans: (d) The vibrations with a constant frequency and constant amplitude

Explanation:

Free vibrations are the vibrations of a body when it is allowed to vibrate on its **own**, without any external periodic force acting on it.

In free vibrations, both the **frequency** and the **amplitude** remain **constant** (ignoring air resistance).

So, option **(d)** is correct.

22.

Ans: (b) Wave *B*

Explanation:

Pitch of a sound depends on its **frequency** – higher frequency means higher pitch.

In the diagram, **wave B** has the **maximum number of oscillations** (more crests and troughs) in the same time compared to A, C and D.

So, wave *B* has the **highest frequency**, and therefore the **highest pitch**.

23.

Ans:

(i) The type of vibration shown in the figure is **damped vibration**.

(ii) The amplitude of vibration decreases with time because a **resistive force (such as friction or air resistance)** opposes the motion.

(iii) An example of natural vibrations is:

A **tuning fork** vibrating freely after being struck.

24.

Ans: (i) As the tuning fork B produces louder sound when placed on Y , then the frequency of tuning fork $E = 320$ Hz.

(ii) B produces louder sound due to the condition which is known as resonance.

In resonance, when the frequency of an externally applied periodic force on a body is equal to its natural frequency, the body readily begins to vibrate with an increased amplitude.

25.

Ans: Pitch of a note is the subjective property of sound that helps us distinguish a **sharp (high-pitch)** sound from a **dull (low-pitch)** sound.

Pitch depends on the **frequency (ν)** of the source.

A note has a **high pitch** when its frequency is high, and a **low pitch** when its frequency is low.

26.

Ans: (i) Amplitude is the maximum displacement of a vibrating particle from its mean position on either side.

It is denoted by **A** and its SI unit is **metre (m)**.

(ii) Frequency is the number of complete vibrations (or waves) produced in one second.

It is denoted by **ν (nu)** and its SI unit is **hertz (Hz)**.

27.

Ans: An echo is not heard when the distance between the source of sound and the reflecting surface is only **10 m** because the sound takes too little time to return.

The sensation of sound remains in our ears for about **0.1 s**.

For an echo to be heard distinctly, the reflected sound must reach the ear **after 0.1 s**, which requires the reflecting surface to be at least **17 m** away.

Since **10 m < 17 m**, the original sound and the reflected sound merge, and no echo is heard.

28.

Ans: (c) Echocardiogram

Explanation:

An **echocardiogram** works using **ultrasound waves**, which reflect off the heart's tissues.

These reflected sound waves form images of the heart.

Therefore, it is a diagnostic method based on **reflection of sound**.

29.

Ans: (c) Ultrasonic waves are used

Explanation:

Echo-depth sounding (used in ships and submarines) works with **ultrasonic waves**.

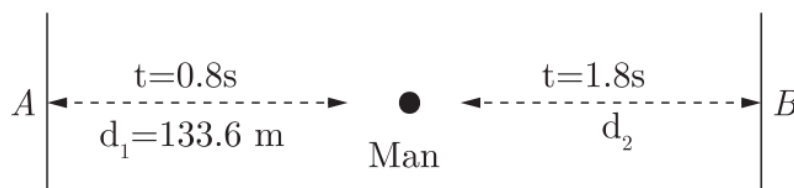
These waves have very high frequency and can travel long distances underwater.

They reflect back after hitting the sea floor or an object, helping to measure **depth and distance**.

30.

Ans: (a) 434.2 m

Explanation:



The man is 133.6 m from the nearer cliff.

For the **first echo** (from the nearer cliff):

Time taken, $t_1 = 0.8\text{ s}$

Distance travelled by sound = 2×133.6 (to cliff and back)

So, speed of sound

$$v = \frac{2 \times 133.6}{0.8} \approx 334 \text{ m s}^{-1}$$

For the **second echo** (from the farther cliff):

Time taken, $t_2 = 1.8 \text{ s}$

Let distance of man from the farther cliff be d_2 .

$$2d_2 = vt_2 = 334 \times 1.8 \Rightarrow d_2 = \frac{334 \times 1.8}{2} = 300.6 \text{ m}$$

Distance between the two cliffs

$$= d_1 + d_2 = 133.6 + 300.6 = 434.2 \text{ m}$$

Hence, the distance between the cliffs is **434.2 m**.

31.

Ans: At a certain speed of the motor bike, the frequency of vibrations produced by the engine becomes equal to the *natural frequency* of the rear-view mirror.

When this happens, **resonance** occurs, causing the mirror to vibrate with a much larger amplitude.

Hence, it starts vibrating violently.

32.

Ans: (i) Given:

$$f = 256 \text{ Hz}, \lambda = 1.3 \text{ m}$$

Speed of sound,

$$v = f\lambda = 256 \times 1.3 = 332.8 \text{ m/s}$$

(ii) If the sound has wavelength 2.6 m but travels with the **same speed**, then:

$$f = \frac{v}{\lambda} = \frac{332.8}{2.6} \approx 128 \text{ Hz}$$

This frequency is **half** of 256 Hz, so the listener will hear a **duller, lower-pitched** sound compared to the original sound.